

Ruptured Ectopic



Section I: Scenario Demographics

Scenario Title:	Ruptured Ectopic		
Date of Development:	27/05/2015 (DD/MM/YYYY)		
Target Learning Group:	☒ Juniors (PGY 1 – 2)	☐ Seniors (PGY ≥ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
Affiliations/Institution(s):	McMaster University
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To ensure that learners can appropriately resuscitate a patient with intra-peritoneal bleeding.
CRM Objectives:	Effectively lead team members through complex critical scenario.
Medical Objectives:	1) Recognize potential ruptured ectopic and initiate resuscitation, workup, and consultation. 2) Administer massive transfusion in appropriate blood product ratio using adjuncts to increase speed of delivery (e.g. –pressure bag, level one). 3) Demonstrate a broad approach to the initial management and workup of a patient presenting with syncope and hypotension

Case Summary: Brief Summary of Case Progression and Major Events

26 year-old female, recently immigrated from Cambodia, presents after a syncopal episode at home. At the case outset, she complains of feeling “a little dizzy” and has a HR of 100 and a BP of 90/60. Once the team initiates care, the patient will say she has to vomit and then become poorly responsive and more hypotensive. The patient does not know that she is pregnant, so the team will have to consider the diagnosis early and use bedside U/S to point them in the right direction. The team will then need to initiate a massive transfusion and arrange for surgery. If the ectopic pregnancy is not recognized, the patient will become persistently more hypotensive until she has a PEA arrest.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.

Massive transfusion: <http://lifeinthefastlane.com/trauma-tribulation-026/>



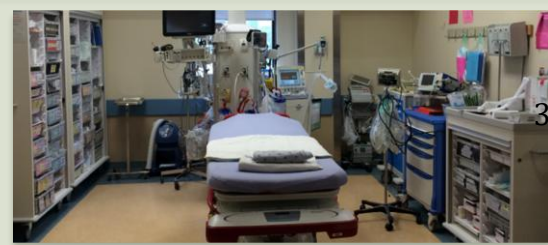
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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates	Brief Description of Role		
None.			
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☒ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☒ Arterial Line Kit	
☒ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
None required.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

26 year old female presents after a syncopal episode at home. She immigrated from Cambodia two weeks ago to work as a live-in nanny, but has been feeling unwell for the last 3 days. The patient speaks limited English, but the family she is staying with said she has been vomiting the past few days and was unable to get out of bed this morning. When she tried, she became quite dizzy and then passed out.

B. Patient Profile and History

Patient Name: Mindy Loewen	Age: 26	Weight: 70kg
Gender: ☐ M ☒ F	Code Status: Full	
Chief Complaint: Syncope		
History of Presenting Illness: Feeling unwell for 3 days. Vomiting and unable to get out of bed this morning. When she tried, she became dizzy and passed out.		
Past Medical History:	None.	Medications: Prenatal vitamins

Allergies: Shellfish

Social History: Lives with family for which she is a live-in nanny.

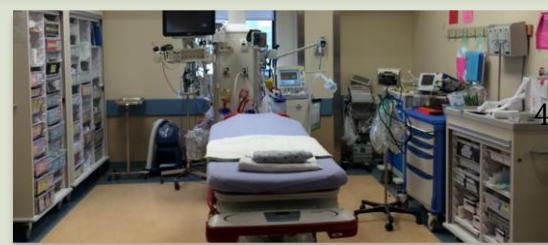
Family History: None significant.

Review of Systems:	CNS:	Feels dizzy and lightheaded. No headache.		
	HEENT:	No visual/speech changes.		
	CVS:	No palpitations.		
	RESP:	No SOB.		
	GI:	Vomiting x3 days. Vague abdominal pain.		
	GU:	No LUTS.		
	MSK:	Nil.	INT:	Nil.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 100/min	BP: 90/60	RR: 12/min
O ₂ SAT: 98%		
Rhythm: NSR	T: 36.5°C	Glucose: 6.3 mmol/L
GCS: 15 (E4 V5 M6)		
General Status: Looks pale. In no distress.		
CNS:	Alert, appropriate. No focal neuro deficits.	
HEENT:	PERLA. 3mm.	
CVS:	No murmur	
RESP:	No advent. GAEB.	
ABDO:	Diffusely tender. No rebound.	
GU:	Nil.	
MSK:	Nil.	SKIN: Pale.

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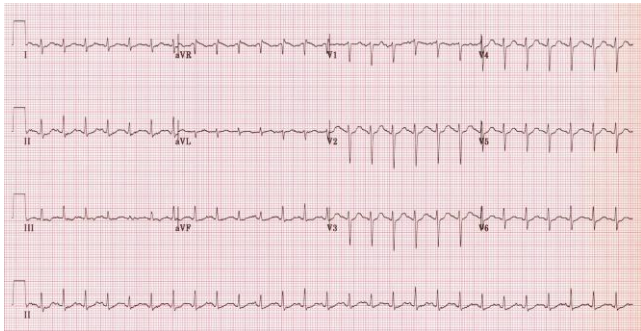
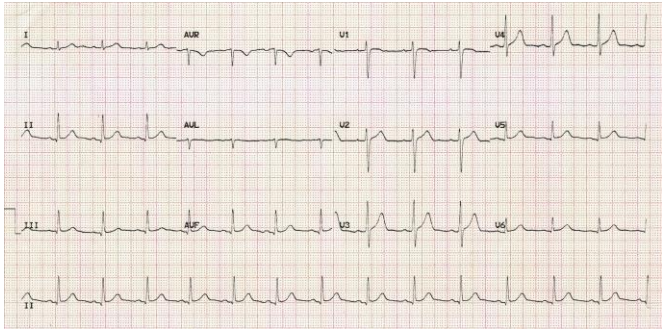
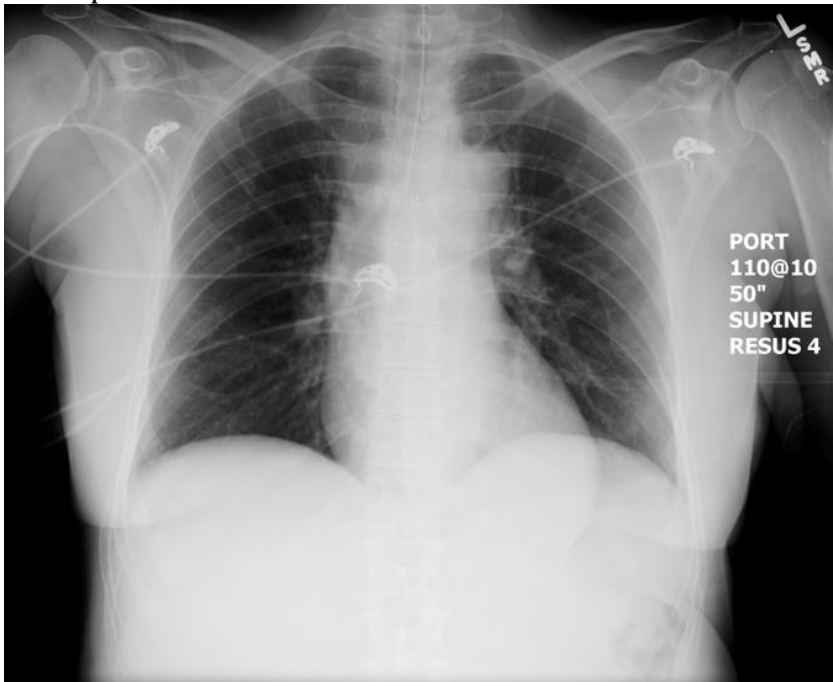
Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: NSR HR: 100/min BP: 90/60 RR: 12/min O ₂ SAT: 98% T: 36.5°C	Pale, but otherwise well. Alert, appropriate.	<u>Learner Actions</u> - ☐ IV, monitors - ☐ Further hx and px - ☐ Cap sugar: 6.3 - ☐ ECG: NSR - ☐ Urine or blood for βHCG - ☐ Basic labs, including G+S - ☐ IV NS 1L bolus	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - 4 min → 2. Syncope
2. Syncope HR → 110 BP → 80/40	Patient starts state by saying “I think I’m going to throw up.” Then is poorly responsive. (Sleepy, groans only, opens eyes to voice.)	<u>Learner Actions</u> - ☐ Establish 2 nd iv (if not yet) - ☐ Bolus IVF x2 - ☐ U/S to r/o ectopic/FF - ☐ 2u PRBC once confirm ectopic - ☐ Call GYNE for OR - ☐ Activate MTP - ☐ ± Intubate for OR	<u>Modifiers</u> - 6 min → BP 70/35, HR 115 - 8 min → BP 65/30, HR 120 - If ask for βHCG result, give back positive result <u>Triggers</u> - If free fluid not identified by 9 min → 3. PEA arrest - Free fluid identified, 4u PRBC given → 4. Resolution
3. PEA Arrest Rhythm → PEA BP → -/-	Patient pulseless and unresponsive.	<u>Learner Actions</u> - ☐ High quality CPR - ☐ Epinephrine q3 min - ☐ Transfuse PRBC/MTP - ☐ ±Intubate	<u>Modifiers</u> - If ectopic not yet ID’d , give urine βHCG saying “the nurse at triage sent it” <u>Triggers</u> - Activate MTP for presumed ectopic → 4. Resolution - 12 min → 4. Resolution
4. Resolution Rhythm → Sinus tach HR → 120 BP → 85/45	Patient intubated & sedated. Stable now. (If not intubated, then awake, but sleepy.)	<u>Learner Actions</u> - ☐ ±Intubate - ☐ Call Gyne for OR - ☐ Continue 1:1:1 transfusion	<u>END CASE → GYNE ARRIVES</u>

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results	
Urine β HCG result given if required.	
Images (ECGs, CXRs, etc.)	
ECG – NSR or sinus tach (depending on state when done)  Sinus tachycardia ECG source: http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/12/sinus-tachycardia.jpg  Normal sinus rhythm ECG source: http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/12/normal-sinus-rhythm.jpg	CXR – post intubation  CXR source: https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg
Ultrasound Video Files (if applicable)	
- FF RUQ	
- No IUP (transabdominal image)	

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) How did it feel to lead the team through a massive transfusion? 2) How do you feel the team communicated about the possible diagnoses and the management plan? 3) If you don't have an official massive transfusion protocol, what is your strategy for transfusion? 4) This patient's presentation was not straightforward. How did you work towards your ultimate diagnosis? Did the team feel like they understood the team leader's thought process?	
Key Moments	
Early differential diagnosis of syncope in young woman → must include β HCG and ECG	
Recognition of ectopic pregnancy as most likely diagnosis (and US to identify it)	
Initiation of massive transfusion and early call for OR.	